FH | JOANNEUM
Elektronik und
Computer Engineering

Radio Frequency Engineering

Laboratory report

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Contents

1 COAXIAL RF LINE

1.1 Task description

This task introduces noise analysis of transistor and operational-amplifier circuits using Cadence PSpice. Following were the main sub-objectives:

- Noise analysis of a single stage emitter circuit
- An OpAmp noise model check (where Gain = +1)
- Noise analysis of inverting and non-inverting amplifier circuits (with Gain = 10)

1.2 Schematics

Figure 1.1 shows the circuit configuration of a single stage emitter circuit.

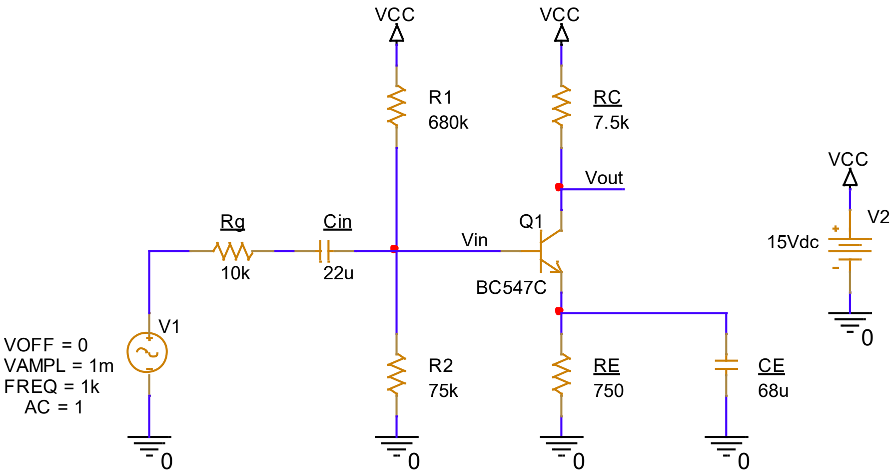

Figure 1.1: PSpice Schematic of a single stage emitter circuit with BC574C.

Figure 1.2 shows the circuit configuration of a non-inverting amplifier circuit.

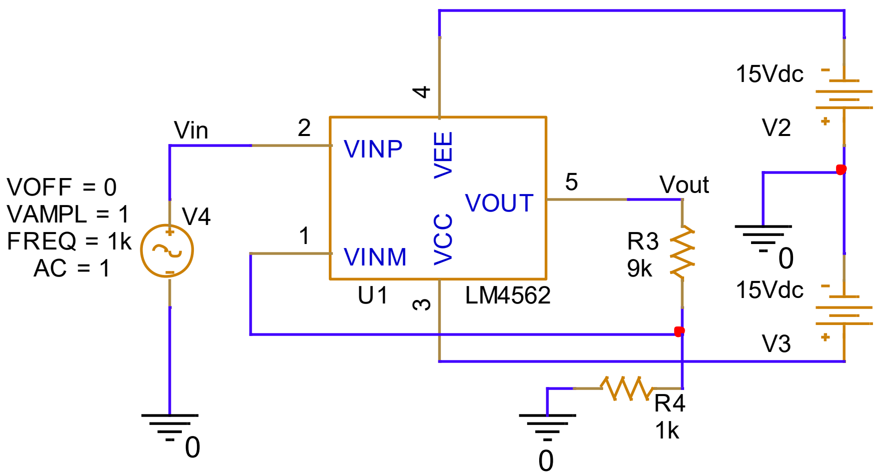

Figure 1.2: Schematic of a non-inverting amplifier with LM4562 (Gain = +10).

Figure 1.3 shows the circuit configuration of an inverting amplifier circuit.

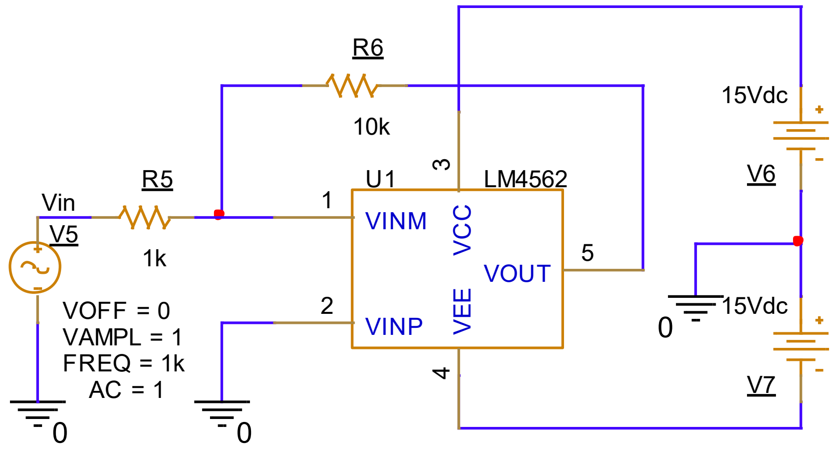

Figure 1.3: Schematic of an inverting amplifier with LM4562 (Gain = -10).

1.3 Table with measurement results

Table 1.1 shows the value of the simulated Input Noise Density and the value in the data sheet for the same.

Adjusted	Measured	Datasheet
f / Hz	e_n / nV/$\sqrt{\text{Hz}}$	e_n / nV/$\sqrt{\text{Hz}}$
10	6.810	6.4 [?]
1.0k	2.752	2.7 [?]

Table 1.1: Values corresponding to the OpAmp noise model check at 10 Hz and 1.0 kHz (Gain = +1).

1.4 Simulation curves and diagrams

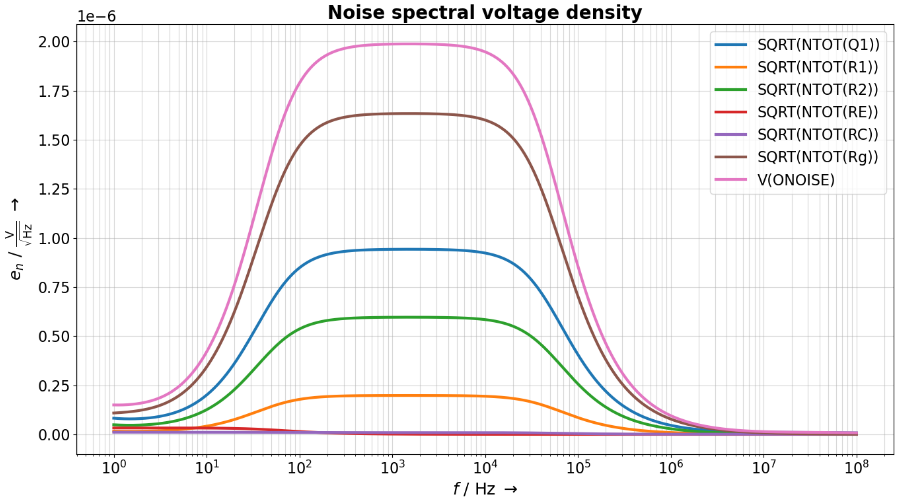

Figure 1.4: Simulation of the noise spectral voltage density of the emitter circuit ($\hat{u}$=1mV).

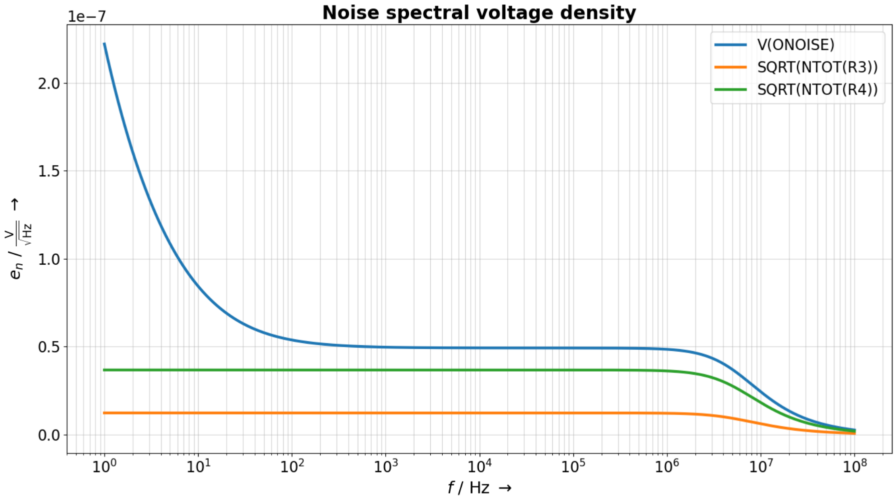

Figure 1.5: Simulation result showing the Noise spectral voltage density of a non-inverting amplifier circuit.

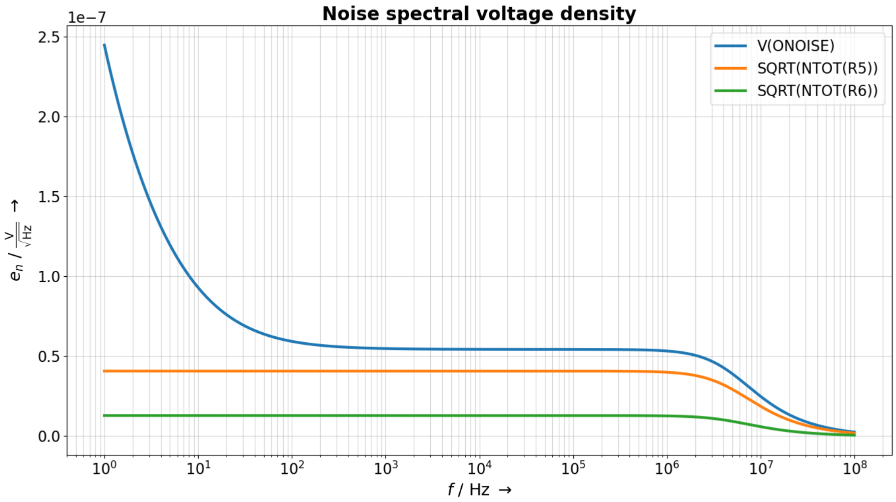

Figure 1.6: Simulation result showing the Noise spectral voltage density of an inverting amplifier circuit.

1.5 Discussion

- **Noise analysis of a single stage emitter circuit**:
  Noise analysis of the single stage emitter circuit (figure 1.4) reveals that R_g contributes most to the total noise (brown trace).

- **OpAmp noise model check**:
  A simple voltage follower circuit was built with LM4562 and using cursors, the simulated noise density e_n was read at 10 Hz as well as 1.0 kHz and compared to the datasheet values (see table 1.1).

- **Noise analysis of inverting and non-inverting amplifier circuits**:
  The simulated output noise spectral density for these circuits with gain-10 amplifiers is shown in (figures 1.5 and 1.6). The spectrum shows the expected behaviour: increased low-frequency noise due to flicker noise and an approximately flat mid-band (white) noise floor. The change in slope marks the 1/f corner frequency.

2 Time Domain Reflectometry

2.1 Measurement A: Rise- and fall time of the line driver

2.1.1 Task Description

The objective of this task is to measure the rise and fall times of a signal generator and the open output of a line driver. The line driver (LM5134 and LMG1025) is supplied with +5V, and the function generator is set to output a rectangular signal with 5Vpp and a 2.5V DC offset.

2.1.2 Schematic

The measurement configuration consists of an Arbitrary Waveform Generator connected to the input of the line driver. The output of the driver is connected to an SMA connector via two 2.7 Ω series resistors, providing a total of 5.4 Ω series resistance. The voltage is detected using a 1 kΩ resistor connected to a second SMA connector, which alongside the 50 Ω input of the oscilloscope and a 50 Ω coaxial transmission line, forms a 1:21 voltage divider.
(Placeholder: Recreate or insert Figure 2-1 from the lab manual showing the block diagram of the measuring arrangement).

2.1.3 Formulas and Calculations

For this specific measurement, the rise and fall times are evaluated directly from the oscilloscope's transient voltage curves using the instrument's automatic measurement functions or cursors (typically measuring the transition from 10% to 90% of the signal amplitude). No additional complex mathematical derivations are required for this step.

2.1.4 Table(s) with measurement results

(Placeholder: Insert table here. Example format below)

Table 2.1: Rise and fall times of the signal generator and line driver.

Adjusted	**Measured**	**Calculated**
Test Point	**Rise Time t_r (ns)**	**Fall Time t_f (ns)**
Signal Generator	[Result]	[Result]
Line Driver (Open Output)	[Result]	[Result]

Notes: Function generator set to 5Vpp, 2.5V DC offset.

2.1.5 Curves & diagrams

(Placeholder: Insert high-resolution vector images of the oscilloscope plots showing the rise and fall times. Ensure cursor values and/or automatic measurement functions are visible in the plot as per guidelines).

2.1.6 Short discussion of measurement results

(Placeholder: Insert a short discussion comparing the rise and fall times of the signal generator versus the line driver. Include an error estimation and explain any deviations from the expected behavior without using general phrases).

2.2 Measurement B: Reflections of a coax cable with different loads

2.2.1 Task Description

An RG142 coaxial cable with a minimum length of 1 m is connected to the line driver output. The task is to terminate the end of the cable with different loads (Open, Short, 50 Ω, 100 Ω, 25 Ω, 10 nF capacitor, and 1 mH inductor) and measure the resulting line reflections. The transient voltage curve must be recorded to determine the time delay between reflection minima, the maximum voltage, and the steady-state asymptotic voltage. Using the measured propagation delay and physical cable length, the relative dielectric constant (ϵ_r) of the cable must be calculated.

2.2.2 Schematic

The basic schematic is identical to Measurement A, but the open end of the RG142 coaxial line is now terminated with various load impedances (Z_L).
(Placeholder: Insert the circuit diagram indicating the transmission line and the variable load attached at the far end).

2.2.3 Formulas and Calculations

The reflection factor ρ_0 at the load is determined by the load impedance Z_L and the characteristic impedance Z_w of the transmission line:

$$\rho_0 = \frac{Z_L - Z_w}{Z_L + Z_w} \tag{2.1}$$

The phase velocity v_p of the signal propagating through the transmission line is related to the speed of light in a vacuum ($c = 2.9979 \times 10^8$ m/s) and the relative dielectric constant ϵ_r:

$$v_p = \frac{c}{\sqrt{\epsilon_r}} \tag{2.2}$$

The time delay t measured between the incident wave and the reflected wave represents the round trip of the signal along the length of the cable l. The delay is:

$$t = \frac{2l}{v_p} \tag{2.3}$$

By substituting the phase velocity formula into the time delay formula, the dielectric constant ϵ_r can be calculated from the measured physical length and measured time delay:

$$\epsilon_r = \left(\frac{2l}{c \cdot t} \right)^2 \tag{2.4}$$

2.2.4 Table(s) with measurement results

(Placeholder: Insert table here. Example format below)

2.2.5 Curves & diagrams

(Placeholder: Insert oscilloscope plots for each of the 7 loads tested. Ensure trace averaging is used, and the time delay between reflection minima is clearly shown using cursors).

Table 2.2: Reflection characteristics and calculated parameters for various loads.

Load	t **(ns)**	V_{max} **(V)**	V_{steady} **(V)**	ρ_0	ϵ_r
Open	[Result]	[Result]	[Result]	[Result]	[Result]
Short	[Result]	[Result]	[Result]	[Result]	[Result]
50 Ω	[Result]	[Result]	[Result]	[Result]	[Result]
100 Ω	[Result]	[Result]	[Result]	[Result]	[Result]
25 Ω	[Result]	[Result]	[Result]	[Result]	[Result]
10 nF	[Result]	[Result]	[Result]	[Result]	[Result]
1 mH	[Result]	[Result]	[Result]	[Result]	[Result]

Notes: Physical length of RG142 cable l = [Measured Length] m.

2.2.6 Short discussion of measurement results

(Placeholder: Analyze the transient responses. Discuss how the different terminations (resistive, capacitive, inductive) affected the reflection coefficient and the steady-state voltage. Provide error estimation for the calculated dielectric constant).

2.3 Measurement C: Determine cable length

2.3.1 Task Description

The goal is to determine the unknown physical length of a provided RG58 coaxial cable. This is achieved using Time Domain Reflectometry by measuring the propagation delay of a reflection and utilizing the relative dielectric constant (ϵ_r) determined in Measurement B.

2.3.2 Schematic

The measurement setup is the same as the previous reflectometry setups, but the standard RG142 cable is replaced with the RG58 coaxial cable of unknown length. The end of the cable is left open or shorted to ensure a strong reflection can be measured.
(Placeholder: Insert a brief schematic or block diagram showing the line driver connected to the "Unknown Length RG58 Cable").

2.3.3 Formulas and Calculations

Using the time delay t measured from the oscilloscope and the phase velocity v_p (derived from the ϵ_r calculated in Measurement B), the unknown length l of the cable can be calculated by rearranging the delay formula:

$$l = \frac{v_p \cdot t}{2} = \frac{c \cdot t}{2\sqrt{\epsilon_r}} \tag{2.5}$$

2.3.4 Table(s) with measurement results

(Placeholder: Insert table here. Example format below)

Table 2.3: Determination of the unknown RG58 cable length.

Assumed ϵ_r	**Time Delay t (ns)**	**Calculated Length l (m)**
Value from B	[Result]	[Result]

2.3.5 Curves & diagrams

(Placeholder: Insert the oscilloscope plot showing the incident and reflected wave used to measure the time delay for the unknown RG58 cable).

2.3.6 Short discussion of measurement results

(Placeholder: Discuss the calculated length. Estimate the uncertainty in the measurement based on the cursor resolution of the oscilloscope and the potential error carried over from the ϵ_r calculated in Measurement B).